

FENGFEI SUN

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EDUCATION

Ph.D. (expected)	Department of Economics, National University of Singapore (NUS)	Aug. 2023 – Present
M.A.	School of Economics, Renmin University of China (RUC)	Sept. 2021 – Jun. 2023
B.A.	School of Economics, Renmin University of China (RUC)	Sept. 2017 – Jun. 2021
Exchange Student, School of Social Sciences, National Taiwan University (NTU), Taiwan		Sept. 2019 – Jan. 2020

RESEARCH INTERESTS

Experimental and Behavioral Economics

Human-AI Interaction, Biases in LLMs, Social Preferences, Public Goods Provision

WORKING PAPERS

Large Language Models are overconfident and amplify human bias

with Goette Lorenz (NUS), Wang Kailong (HUST), and Li Ningke (HUST)

under review

Large language models (LLMs) are revolutionizing every aspect of society. They are increasingly used in problem-solving tasks to substitute human assessment and reasoning. LLMs are trained on what humans write and are thus exposed to human bias. We evaluate whether LLMs inherit one of the most widespread human biases: overconfidence. We algorithmically construct reasoning problems with known ground truths. We prompt LLMs to answer these problems and assess the confidence in their answers, closely following similar protocols in human experiments. We find that all five LLMs we study are overconfident: they overestimate the probability that their answer is correct between 20% and 60%. Humans have accuracy similar to the more advanced LLMs, but far lower overconfidence. Although humans and LLMs are similarly biased in questions which they are certain they answered correctly, a key difference emerges between them: LLM bias increases sharply relative to humans if they become less sure that their answers are correct. We also show that LLM input has ambiguous effects on human decision making: LLM input leads to an increase in the accuracy, but it more than doubles the extent of overconfidence in the answers.

When the Wealthy Step Up: Endowment Heterogeneity and Thresholds in Public Goods

with Lian Hongquan (SCNU), Huang Chunchun (CAU), and Zhou Ye'an (RUC)

in submission

Ensuring successful collective action in the presence of wealth disparities remains a central challenge for policy and institutional design. This study examines how external threshold requirements and heterogeneous endowment structures jointly affect individual performance and cooperation patterns in threshold public goods games under random matching. We find that both thresholds and endowment inequality enhance cooperation, with higher thresholds and greater disparities producing complementary effects. Specifically, at low thresholds, the provision point functions as a salient reference that facilitates basic cooperation, yet conflicting fairness norms emerge under heterogeneous endowments. At high thresholds, however, wealthier participants play a pivotal role, leading to higher success rates. Beliefs about others' contributions and inequality aversion preferences significantly moderate these interactions: high-endowment participants with pronounced disadvantageous inequality aversion tend to increase contributions to move closer to equitable outcomes. Our findings contribute to policy debates by illuminating how provision difficulty, wealth dispersion, and social preferences intersect to shape cooperation in diverse collective-action settings.

Understanding Effects of Decision Recommendation: Evidence from Large Language Models

with Wang Ao (NUS)

in draft

Why Peers Matter: How Friendship Network Affects Students' Academic Performance, *independent author*

2020

CONFERENCES PRESENTATIONS

Annual Conference of the National Game Theory and Experimental Economics Research Association, Shandong University	2025
2025 ESA World Meeting, Renmin University of China	2025
China Behavioral and Experimental Economics Forum, Wuhan University	2023
2023 Asia-Pacific ESA Meeting, Seoul, Republic of Korea	2023
Fifth Prospective Economists Forum (PEF), Central University of Finance and Economics & China Economics Annual Conference & Young Economist Society, the third Prize in graguate group (4 / 612)	2022
Fourth Brain-Like Economics Seminar, Shandong University	2022

LEADERSHIP ACTIVITIES

Co-Organizer of Multi-Universities Online Biweekly Seminar	Apr.2019 – Present
- Organized and coordinated 60+ tutors and students from 10+ universities to hold 50+ seminars. - Reported and reviewed 5 papers from top journals in seminars by now.	
Co-Founder of Economic Knowledge Sharing Platform <i>Beyond Economics</i>	Apr. 2018 – Present
- Accumulated 5,000+ followers on WeChat and expanded into an academic group with 50+ participants. - Released 2 reviews of classic dissertations and 15+ summative articles for frontier papers online.	
Main Leader of Volunteer Service Group <i>The Dreamer</i> in Shandong Province	Nov.2017 – Jun. 2021
- Established and lead non-profit education support organization <i>The Dreamer</i> in the hometown to help high school students in the countryside broaden their horizons and pursue a better education. - Implemented 8 sessions of supporting teaching activities during vacations. 300+ volunteers and 800+ students participated accumulatively. The organization was awarded <i>the Advanced Volunteer Group</i> by the local government in 2019.	

SELECTED AWARDS & HONORS

• Excellent Graduate thesis of Graduation Design , Beijing (1/60)	2024
• Winner in Top-notch Innovative Academic Competition , RUC (Top 5% of candidates)	2022
• Excellent Undergraduate thesis of Graduation Design , Beijing (Top 0.1% of graduates in Beijing)	2021
• Distinguished Undergraduate , RUC (Top 4% of graduates in RUC)	2021
• Outstanding Winner in Extracurricular Academic Competition, RUC (Top 0.5% of participants)	2020
• First-Class Academic Excellence Scholarship , RUC (Top 2% of candidates)	2022, 2020
• National Encouragement Scholarship (ranked 1 st out of candidates)	2019, 2018
• Ranked 1st out of 7500 students in the National College Entrance Exam in hometown, Shandong Province	2017

OTHERS

- Skills:** Proficient in **R**, **STATA**, **Python**, and **LATEX**
- Languages:** Mandarin Chinese (native), English (proficient)